

DUNKIRK AP, NY, USAWMO: **744989**Lat: **42.493N**Lon: **79.272W**Elev: **666**StdP: **14.35**Time Zone: **-5.00 (NAE)**Period: **99-19**WBAN: **14747****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	8.5	-4.7	4.4	4.9	-0.2	5.6	10.0	29.7	36.9	26.9	33.3	8.0	210	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	86.1	72.3	83.8	71.2	81.7	70.0	74.8	82.4	73.3	80.5	72.0	79.2	11.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	122.3	79.7	70.8	116.2	78.2	69.6	111.7	77.2	38.6	82.7	37.2	80.6	36.1	79.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.5	22.4	19.8	DB	-4.6	90.6	9.1	2.0	-11.2	92.0	-16.5	93.2	-21.7	94.3	-28.3	95.7
			WB	-5.0	77.1	8.7	1.4	-11.2	78.1	-16.4	78.9	-21.3	79.7	-27.6	80.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	27.1	27.2	34.4	45.8	57.6	66.7	71.2	69.9	63.7	52.8	42.5	33.0	(6)
	DBStd	18.08	11.47	11.40	10.90	9.33	8.58	6.78	5.34	5.12	6.97	8.27	8.95	8.89	(7)
	HDD50	2921	715	640	498	190	24	0	0	0	1	64	260	529	(8)
	HDD65	6263	1176	1057	948	578	262	60	9	14	106	388	674	991	(9)
	CDD50	2721	3	3	15	65	260	500	656	617	411	151	36	4	(10)
	CDD65	589	0	0	0	4	33	110	200	166	66	10	0	0	(11)
	CDH74	4115	0	0	1	38	243	807	1491	1055	432	48	0	0	(12)
	CDH80	835	0	0	0	3	39	179	331	196	83	4	0	0	(13)
(14) Wind	WSAvg	9.4	11.5	10.8	9.8	9.8	8.4	8.1	7.7	7.5	7.9	9.7	10.9	11.3	(14)
(15) Precipitation	PrecAvg	42.1	2.6	2.0	2.6	3.4	3.4	3.7	3.8	3.9	4.9	4.1	4.4	3.3	(15)
	PrecMax	58.1	5.6	5.2	5.4	6.4	6.8	9.8	9.2	9.0	11.9	7.4	12.1	7.0	(16)
	PrecMin	31.4	0.9	0.4	0.6	0.9	0.9	1.1	1.1	0.9	2.1	0.9	1.5	1.3	(17)
	PrecStd	6.3	1.1	1.1	1.1	1.3	1.5	1.8	1.9	1.7	2.2	1.4	2.0	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	61.4	69.3	79.2	84.2	88.0	89.2	88.1	86.5	79.4	68.7	62.8	(19)
		MCWB	54.8	52.3	58.4	62.2	70.0	71.1	74.3	74.0	72.4	67.2	57.4	55.5	(20)
	2%	DB	54.5	54.5	62.6	72.9	80.3	84.4	86.1	84.4	82.2	74.0	64.8	54.7	(21)
		MCWB	49.5	48.8	53.2	58.5	66.5	70.8	72.8	71.9	70.5	64.1	55.5	48.9	(22)
	5%	DB	48.1	48.4	57.2	67.7	76.3	81.8	83.7	81.9	79.1	70.2	61.4	50.6	(23)
		MCWB	43.0	43.2	49.6	55.6	64.4	69.3	71.5	71.1	68.8	62.4	53.5	46.4	(24)
	10%	DB	43.0	42.4	51.7	62.5	72.5	79.0	81.4	79.7	75.5	66.0	57.0	45.9	(25)
		MCWB	39.0	37.6	45.3	52.7	62.0	67.9	70.2	69.4	66.8	58.7	50.1	41.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	55.6	59.0	65.0	71.4	74.4	77.1	76.3	74.9	69.5	60.3	55.8	(27)
		MCDB	61.0	60.8	67.4	75.4	82.1	83.6	85.9	84.5	82.6	76.0	65.3	61.9	(28)
	2%	WB	50.1	49.2	54.6	60.3	68.7	72.4	75.2	74.3	72.6	66.8	57.5	50.8	(29)
		MCDB	53.7	54.3	60.8	70.3	77.7	81.3	83.1	81.2	79.6	72.3	63.1	54.4	(30)
	5%	WB	44.0	43.8	50.4	57.3	66.4	71.0	73.5	72.9	70.3	63.0	54.2	46.2	(31)
		MCDB	47.5	48.0	56.7	65.6	74.4	79.5	80.6	79.5	76.8	68.3	60.2	49.8	(32)
	10%	WB	38.7	37.4	45.5	53.6	63.9	69.3	72.1	71.3	68.3	59.8	51.0	42.1	(33)
		MCDB	42.8	41.8	51.1	60.6	70.6	77.2	79.1	77.8	74.3	65.4	56.5	45.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	14.2	15.5	18.5	18.7	16.9	16.3	16.2	16.9	14.9	13.5	11.2	(35)
		MCDBR	18.0	21.8	23.2	26.5	22.0	19.9	17.8	18.2	19.7	20.0	19.7	17.9	(36)
	5% WB	MCWBR	15.3	17.7	16.1	15.9	11.9	10.5	9.3	10.4	11.2	12.2	14.1	15.0	(37)
		MCWBR	18.2	21.6	22.0	23.9	19.7	16.8	15.0	14.9	16.6	17.7	18.2	16.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.305	0.340	0.337	0.373	0.404	0.435	0.438	0.428	0.378	0.361	0.315	0.303	(40)
		taud	2.388	2.258	2.368	2.328	2.288	2.235	2.249	2.267	2.379	2.408	2.506	2.455	(41)
	Ebn at Noon		264	269	288	284	276	267	265	264	270	260	261	255	(42)
		Edh at Noon	25	34	34	38	41	43	43	41	34	29	23	22	(43)
(44) All-Sky Solar Radiation	RadAvg		407	670	1008	1396	1717	1866	1882	1672	1318	803	493	312	(44)
	RadStd		38	68	128	148	175	141	144	94	124	81	75	44	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page